

A Systematic Replicability Study of BSARec for Sequential Recommendation

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*Equal Contribution

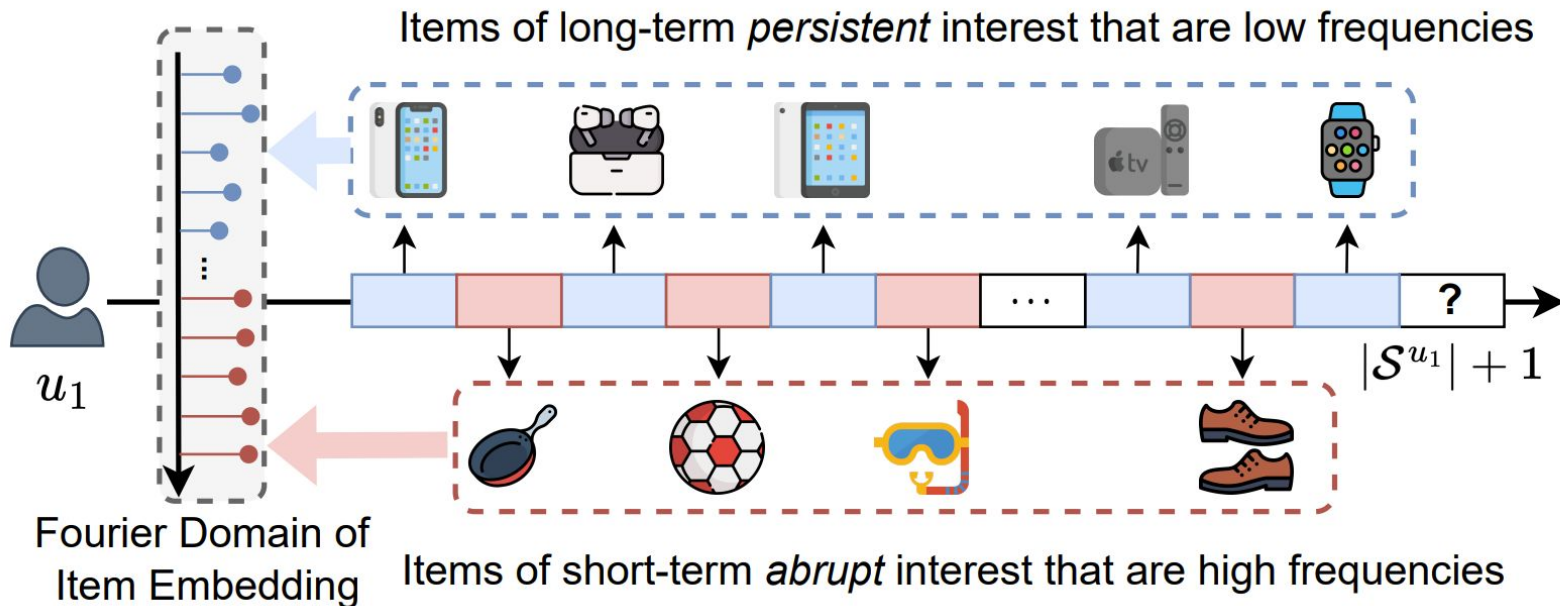


ECIR 2026, Delft - The Netherlands

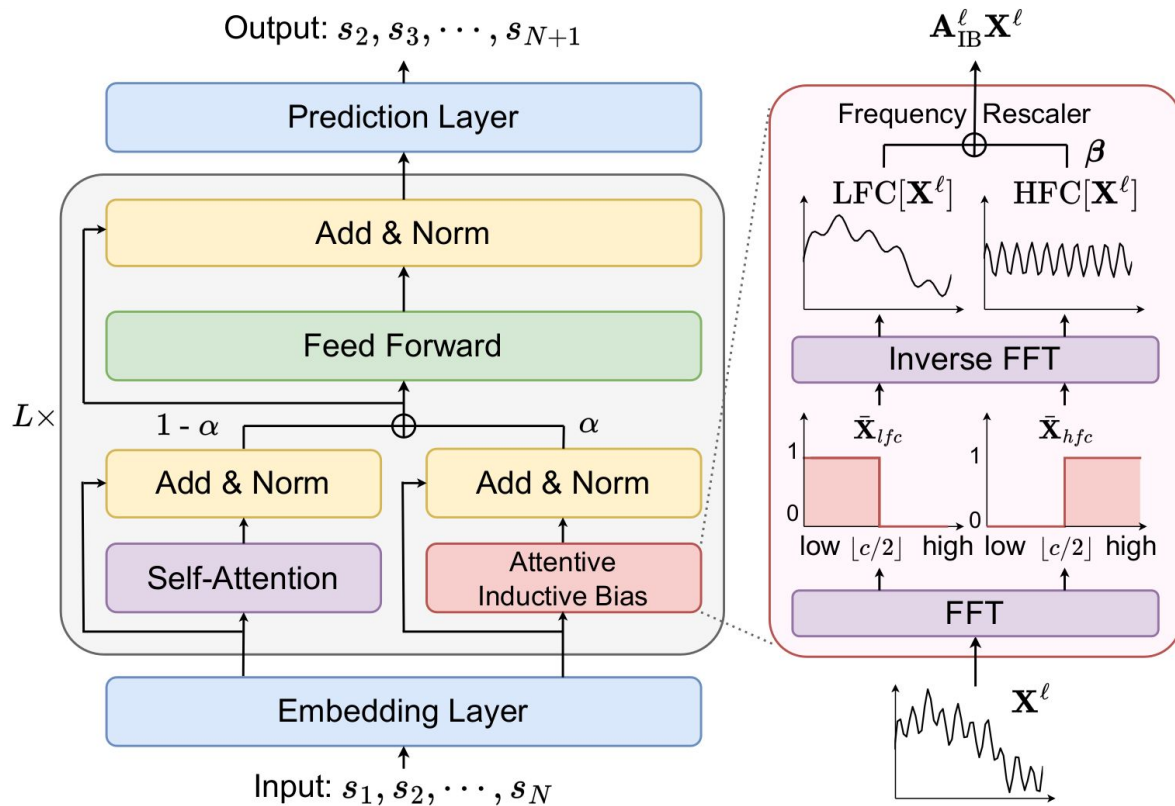
BSARec: Short-Term Inductive Bias

Users have both long-term and short term interest. Can we account for this explicitly in Sequential Recommendation?

Core Idea



BSARec Layer



Motivations: BSARec Main Claim

- Can we **reproduce** positive results of BSARec?
- **Why** does the attentive inductive bias work?
 - Are high-frequency signals **effectively** captured?
 - Is the **Fourier transform** suitable for **SR signals**?
 - How is BSARec **affected** by different **sequence lengths**?

Results

- Statistically significant **performance increase** on **5/7 datasets**
- **Performs equally** as well as the **best baseline** on ML-1M
- **Worse** on the MIND dataset

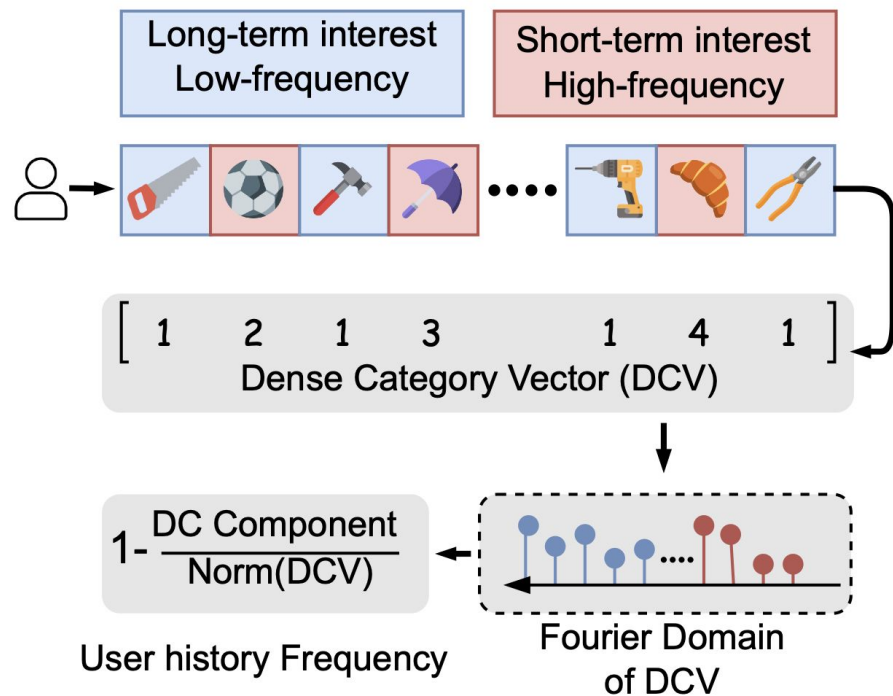
Dataset	Metric	SASRec	BERT4Rec	FMLPRec	duoRec	FeaRec	BSARec	Diff.
ML-1M	HR@10	0.2008	0.2234	0.2090	0.2678	<u>0.2700</u>	0.2746	1.73
	NDCG@10	0.1054	0.1220	0.1085	<u>0.1512</u>	0.1503	0.1539	1.80*
LastFM	HR@10	0.0546	0.0495	<u>0.0591</u>	0.0547	0.0543	0.0703	18.94*
	NDCG@10	0.0303	0.0256	<u>0.0316</u>	<u>0.0338</u>	0.0331	0.0405	19.73*
MIND	HR@10	0.1068	0.1739	0.0987	0.1692	<u>0.1710</u>	0.1633	-6.11
	NDCG@10	0.0584	0.0947	0.0529	0.0926	<u>0.0943</u>	0.0899	-5.07

Table 1. Results on several SR benchmarks.

RQ1 Performance of SR with high and low frequencies

How to Measure frequencies

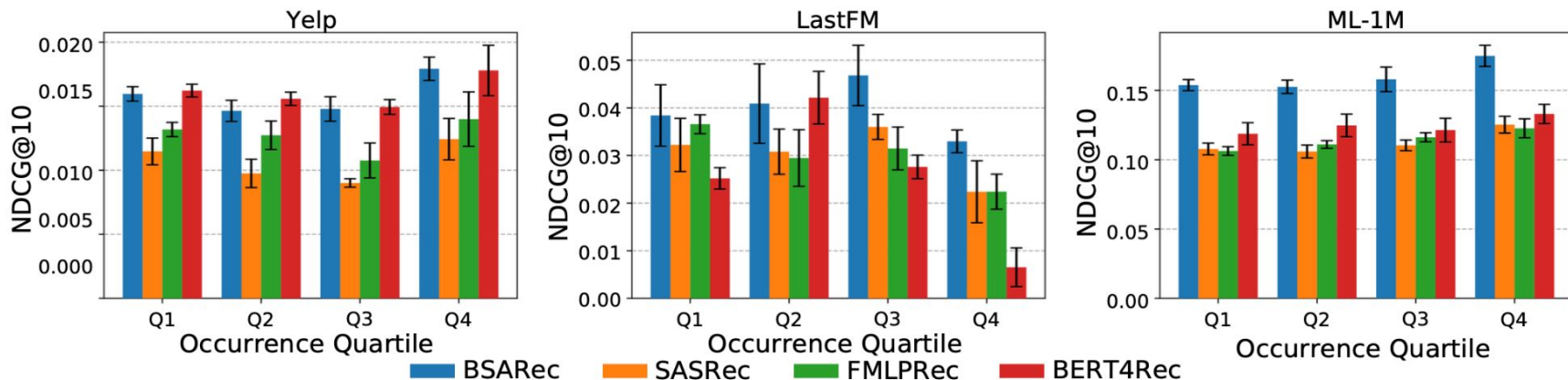
- Evaluate model performance for sequences with **high-frequency** items
- How can we **group** items into high and low-frequencies?



RQ1 Performance of SR with high and low frequencies

Results

- BSARec shows **significantly better** performance over other methods on LastFM and ML-1M for **high frequency-groups**
- Difference in performance** between quartiles is **dataset-dependent**.

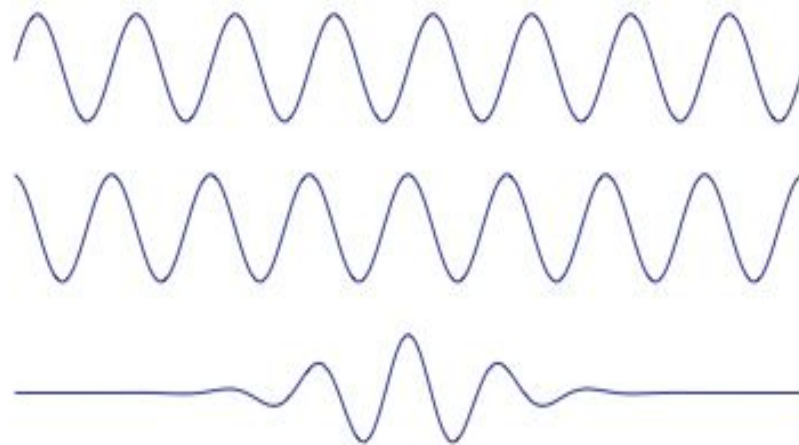


Locality of User Interactions

Fourier Transform



Wavelet Transform

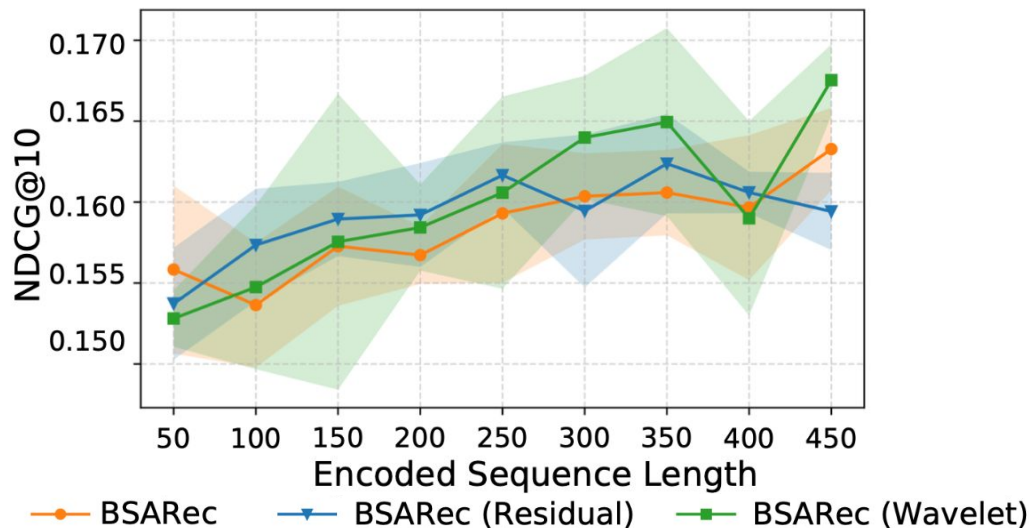


Wavelet transform in SR potentially allows for improved performance over Fourier transform due to locality (Lu et al.,2025).

Core Idea

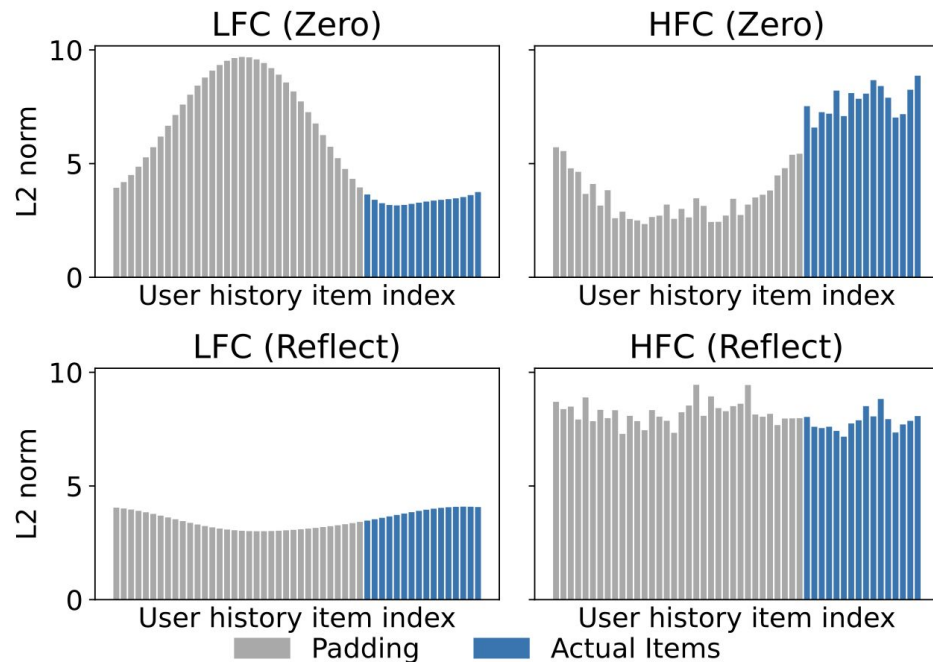
Results

- **Residual connection** Baseline can **outperform BSARec** for sequence lengths up to 250.
- **Wavelet transform** shows **marginal** performance **improvements**.



Padding matters

- The **frequency component** of our signal is affected by **padding**
- **Zero padding** is processed as **low-frequency signal**, distorts frequencies.



Results

- **Non-zero padding** leads to **significant** performance **improvements**.
- Effect is **proportional** to amount of padding **added**.

Dataset	Metric	Zero	Cyclic	Reflect	Symmetric	Diff.
LastFM	HR@5	0.0497	0.0479	0.0499	0.0490	-0.37
	HR@10	0.0686	0.0730	0.0760	0.0701	-9.66*
	HR@20	0.1017	0.1073	0.1114	0.1072	-8.73*
	NDCG@5	0.0347	0.0329	0.0345	0.0346	0.42
	NDCG@10	0.0408	0.0411	0.0430	0.0414	-5.15
	NDCG@20	0.0490	0.0497	0.0518	0.0507	-5.47*
ML-1M	HR@5	0.1892	0.1946	0.1947	0.1945	-2.82
	HR@10	0.2784	0.2807	0.2817	0.2801	-1.14
	HR@20	0.3832	0.3864	0.3882	0.3860	-1.27
	NDCG@5	0.1271	0.1317	0.1315	0.1306	-3.44
	NDCG@10	0.1558	0.1594	0.1595	0.1582	-2.28
	NDCG@20	0.1822	0.1860	0.1863	0.1849	-2.18
Yelp	HR@5	0.0178	0.0187	0.0193	0.0190	-7.43*
	HR@10	0.0310	0.0323	0.0332	0.0328	-6.58*
	HR@20	0.0513	0.0548	0.0556	0.0554	-7.77*
	NDCG@5	0.0112	0.0117	0.0121	0.0119	-7.79*
	NDCG@10	0.0154	0.0161	0.0166	0.0164	-7.22*
	NDCG@20	0.0205	0.0217	0.0222	0.0220	-7.77*

Contributions/ Conclusions

REP

Successfully reproduced main results shown in Shin et al. (2024)
Significantly worse results on added MIND dataset.

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Using proposed Scaled DC component metric, we can measure and compare models based on frequency-components
BSARec can outperform other models, differences are dataset-specific

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Using proposed Scaled DC component metric, we can measure and compare models based on frequency-components
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RQ2

The Choice of Discrete Signal Processing method is relevant for BSARec.
Differences between methods and residual connections are marginal.

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Using proposed Scaled DC component metric, we can measure and compare models based on frequency-components
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RQ2

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Differences between methods and residual connections are marginal.

RQ3

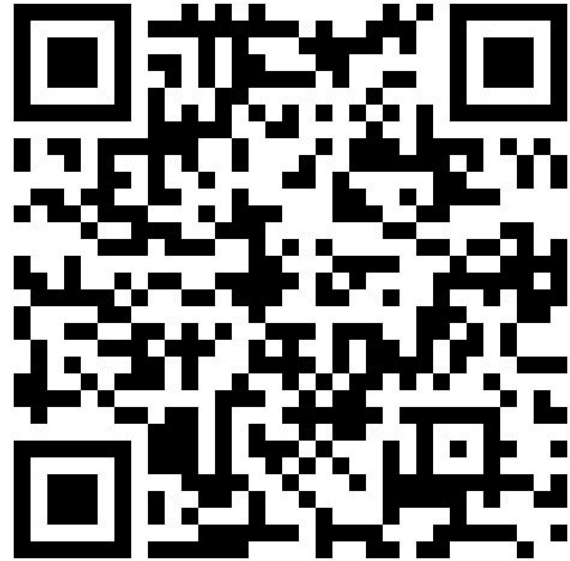
Padding user histories with Zeroes distorts low-frequency signal
By applying other methods of padding, we can improve performance significantly in high-padding settings.

Q&A

Paper



GitHub



Sources

Images

- User-item interest illustration is from Shin et al. (2024). An Attentive Inductive Bias for Sequential Recommendation beyond the Self-Attention
- Wavelet image is taken from <https://blog.wolfram.com/2011/10/26/wavelets-and-their-application-in-mathematica/>

Literature

- Shin, Y., Choi, J., Wi, H., & Park, N. (2024). An Attentive Inductive Bias for Sequential Recommendation beyond the Self-Attention. Proceedings of the AAAI Conference on Artificial Intelligence, 38(8), 8984-8992. <https://doi.org/10.1609/aaai.v38i8.28747>
- Lu et al. (2025). Filtering with Time-frequency Analysis: An Adaptive and Lightweight Model for Sequential Recommender Systems Based on Discrete Wavelet Transform
- Zihua Si, Zhongxiang Sun, Jiale Chen, Guozhang Chen, Xiaoxue Zang, Kai Zheng, Yang Song, Xiao Zhang, Jun Xu, and Kun Gai. 2024. Generative Retrieval with Semantic Tree-Structured Identifiers and Contrastive Learning. In Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP 2024). Association for Computing Machinery, New York, NY, USA, 154–163. <https://doi.org/10.1145/3673791.3698408>
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